

XIN YAO

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RESEARCH INTERESTS

My research lies at the intersection of machine learning, wireless communication systems, and information theory. I combine theoretical analysis with algorithm and system design to develop efficient, secure, and adaptive methods for distributed learning and dynamic network control, and evaluate them on real-world wireless testbeds that I build from the physical layer up.

EDUCATION

University of Florida, Gainesville, FL, USA Ph.D. Student, Electrical and Computer Engineering Advisor: Dr. Mingyue Ji	08/2024 – Present
University of Utah, Salt Lake City, UT, USA Ph.D. Studies, Electrical and Computer Engineering Advisor: Dr. Mingyue Ji	08/2021 – 07/2024
George Washington University, Washington, DC, USA M.S., Electrical Engineering	01/2020 – 06/2021
Tianjin University of Science and Technology (TUST), Tianjin, China B.Eng., Electronic Information Engineering	09/2015 – 06/2019

RESEARCH EXPERIENCE

UAV-Based Wireless Communication Platform

- Assemble customized drones and integrate wireless communication payloads for airborne networking experiments. Initialize and tune flight controllers; design and mount customized communication payloads.
- Design the protocol between devices on the floor and the one on the air.

Unionlabs: AWS-based Remote Access and Sharing of NextG and IoT Testbeds

- Develop and deploy a web user interface on AWS for remote access to the wireless testbed.
- Design the different physical layer (PHY) codes for different hardware including USRP and LoRa.
- Design the unified Unionlabs applications (union API) to read and use different PHY codes, and connect the PHY codes and users' algorithm automatically.
- Design the abstract layer for users to use the PHY and union API with single command.
- Assist the collaborators from different universities to apply their learning or optimizing algorithm using Unionlabs. Users are able to use multiple testbeds at the same time.

EdgeAI Lab: Real-World Wireless Communication Testbed

- Implement the physical-layer software stack in C++: modulation, demodulation, filtering, synchronization, channel encoding/decoding, phase offset and carrier frequency offset.
- Build Python APIs on top of the C++ core to enable scripted, repeatable over-the-air experiments.
- Design automated experiment workflows that adapt to heterogeneous physical-layer hardware.
- Apply the our group's different learning algorithms on lab-wise and building-wise testbeds.

Constrained optimization for real-world problem using Reinforcement learning

- Derive the optimization objective using Lyapunov functions with real-world constraints.
- Design novel state representations for RL algorithms applied to communication-system control.
- Evaluate dynamic-control policies in simulation and deployment on EdgeAI Lab testbed across multiple reinforcement-learning algorithms.

Network optimization with low power (LoRa) communication

- Derive the optimization objective using Lyapunov functions with real-world constraints.
- Design the system architecture for adapting of LoRa PHY parameters jointly.
- Based on the jointly adapt method, design the path optimize and discover system to guarantee the always-live communication against the interference and hardware outages.
- Evaluate dynamic-control policies in simulation and deployment on EdgeAI Lab testbed across multiple reinforcement-learning algorithms.

A-DEFEAT: adaptive error-feedback mechanism under decentralized learning

- Designed a new decentralized learning algorithm with adaptive error-compensation mechanism.
- Propose key proof technique for adaptive error-feedback mechanism. Convergence analysis of proposed learning algorithm and compare convergence rate with existing algorithms.
- Evaluation the proposed algorithm and compare with existing algorithms under different settings.

Information-Theoretic Secure Aggregation with Groupwise Keys

- Studied the information-theoretic foundations of secure aggregation for distributed learning.
- Implemented and compared the proposed secure-aggregation scheme and other existing algorithms.

Master Projects

George Washington University

- **Age of Information with Energy Harvesting** (Advisor: Prof. Omur Ozel): Analyzed age of information (AoI) in M/G/1/1 and idealized infinite-buffer queueing models, including variants with packet deadlines; Investigated AoI-minimization strategies for the bufferless M/G/1/1 model with packet deadlines.
- **Secure File Transfer: Encryption and Decryption for FTP** (Advisor: Prof. Tian Lan): Designed a custom encryption/decryption scheme for FTP on Linux, with client-side encryption and server-side decryption of commands and data; Verified end-to-end FTP transmission and the encrypted.

Undergraduate Projects

Tianjin University of Science and Technology

- **Voice-controlled smart-home system** (Advisor: Jing He): designed STM32F103-based hardware with an LD3320 voice-recognition module; wrote firmware linking the microcontroller to speaker, PC, and LCD peripherals; tested spoken commands with system feedback.
- **License-plate recognition system** (Advisor: Prof. Yang An): built a MATLAB pipeline for image capture from surveillance cameras, filtering (Hamming window), and character segmentation and matching.
- **Android music player** (Advisor: Prof. Ya-Jun Li): developed a multi-format music player in Java using Eclipse and the Android SDK.
- **Digital counter** (Advisor: Prof. Yu-liang Liu): designed the counter circuit, selected components, and assembled, tested, and debugged the board.

PUBLICATIONS

Conference Papers

1. **X. Yao**, A. Bhuyan, and M. Ji, "FORGE-LoRa: Joint per-Slot PHY Adaptation for Multi-Hop LoRa Networks", National Capital Region, USA, 2026 (accepted).

2. **X. Yao**, X. Wang, H. Zhang, and M. Ji, "Experimental Evaluation of Reinforcement Learning for Spectrum Sharing on a Real-Time LoRa Testbed," in *Proc. IEEE International Symposium on Spectrum Innovation (DySPAN)*, Washington, DC, USA, 2026, pp. 1–8.
3. **X. Yao**, A. Bhuyan, X. Zhang, and M. Ji, "A Novel LDPP-MADDPG Approach for Distributed Power Allocation in mmWave Cellular Networks," in *Proc. IEEE Military Communications Conference (MILCOM) — 6G and Spectrum Security for Critical Communication (6GSECC) Workshop*, 2025.
4. K. Wan, **X. Yao**, H. Sun, M. Ji, and G. Caire, "GroupSecAgg: Information Theoretic Secure Aggregation with Uncoded Groupwise Key," in *Proc. IEEE International Conference on Communications (ICC)*, Rome, Italy, 2023, pp. 3890–3895.

Journal Articles

1. K. Wan, **X. Yao**, H. Sun, M. Ji, and G. Caire, "On the Information Theoretic Secure Aggregation with Uncoded Groupwise Keys," *IEEE Transactions on Information Theory*, vol. 70, no. 9, pp. 6596–6619, Sept. 2024.

PRESENTATIONS, POSTERS AND DEMOS

Presentations

1. "FORGE-LoRa: Joint per-Slot PHY Adaptation for Multi-Hop LoRa Networks", National Capital Region, USA, 2026 (upcoming).
2. "Experimental Evaluation of Reinforcement Learning for Spectrum Sharing on a Real-Time LoRa Testbed," in *Proc. IEEE International Symposium on Spectrum Innovation (DySPAN)*, Washington, DC, USA, 2026.
3. "A Novel LDPP-MADDPG Approach for Distributed Power Allocation in mmWave Cellular Networks," in *Proc. IEEE Military Communications Conference (MILCOM) — 6G and Spectrum Security for Critical Communication (6GSECC) Workshop*, 2025.
4. "GroupSecAgg: Information Theoretic Secure Aggregation with Uncoded Groupwise Key," in *Proc. IEEE International Conference on Communications (ICC)*, Rome, Italy, 2023.
5. "High-Band Frequency Spectrum for 5G" (GWU team presentation), DC, USA, 2020.

Posters and demos

1. "A-DEFEAT: An Adaptively Discounted Error-Feedback Algorithm for Decentralized Learning" (poster), North American School of Information Theory (NASIT) BYU, Provo, UT, USA, Jun. 2026.
2. "OSWireless: automatic system to maximize the network throughputs" (demo), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2025.
3. "UnionLabs: Shared Access to heterogeneous Wireless IoT and Wireless OS Testbeds" (poster), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2025.
4. "MADDPG based Power Allocation Scheme using Measurement Data from POWDER" (poster), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2024.

TECHNICAL SKILLS

- **Programming:** Python, C/C++, Java, JavaScript, MATLAB, Arduino, JSON, XHTML/DHTML
- **Machine learning:** reinforcement learning, federated learning, secure aggregation, distributed and parallel computing, centralized learning, decentralized learning
- **Wireless and signal processing:** physical-layer implementation (modulation/demodulation, synchronization, filtering, channel encoding/decoding), real-world testbed development, network protocol analysis

- **Embedded and cloud systems:** STM32 microcontrollers, Android SDK (Eclipse/JDK), AWS, Linux
- **Design and productivity tools:** AutoCAD, Siemens NX, Sharp3D, Adobe (Photoshop, Premiere Pro, After Effects, Acrobat), MS Office

HONORS AND AWARDS

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| • Third-Prize Scholarship, TUST | 2017 |
| • Advanced Individual of Social Work, TUST | 2016 |
| • Excellent League Member, TUST | 2016 |